Bangalore Institute of Technology

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Department of Electronics Engineering (VDT)

MINI-PROJECT BVL586 PRESENTATION

Solar Wireless Electric Vehicle Charging System

Under the guidance of:

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INSTITUTE VISION AND MISSION

Vision: To establish and develop the institute as the center of higher learning, ever abreast with expanding horizon of knowledge in the field of engineering and technology, with entrepreneurial thinking, leadership excellence for life-long success and solve societal problem

Mission:

1. Provide high quality education in the engineering disciplines from the undergraduate through doctoral levels with creative academic and Professional programs.
2. Develop the Institute as a leader in Science, Engineering, Technology and management, Research and apply knowledge for the benefit of society.
3. Establish mutual beneficial partnerships with industry, alumni , local , state and central governments by public service assistance and collaborative research.

DEPARTMENT VISION , MISSION, PEOs, PSOs

PEOs

1. Prepare graduates to be professionals, practicing engineers entrepreneurs in the field of electronics and communication
2. To acquire sufficient knowledge base for Innovative Techniques in Design and development of tools and systems

PSOs

1. Capable of competing in Globally in **Multidisciplinary field.**
2. Achieve personal and professional success with awareness and commitment to ethical and **social responsibilities** as an individual as well as team.
3. Graduates will maintain and improve technical competence through continuous **learning process**

Course Outcomes of Mini-Project

1	Ability to carry-out literature review and define the problem.
2	Ability to co-ordinate to work as a team member or a single member.
3	Inculcate methods to use advance tools .
4	Design analytical modeling and develop a systems
5	Ability to equip analysis skills and interpretation.
6	Enhance presentation skills, drafting and documentation of project work.

CO-PO Mapping Table

PO / CO	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2	PSO 3
1	3	2	2			1		1	1	1			2	2	
2	3	2	2			1		1	1	1			2	2	
3	3	2	2		3			1	1	1			2	2	
4	3	2	2		3	1		1	1	1			2	2	
5	3	2	2		3	1		1	1	1			2	2	
6	3				3			1	1	1	2		2	2	

ABSTRACT

- The increasing adoption of electric vehicles (EVs) is limited by the lack of sufficient charging stations. This research introduces a wireless EV charging system powered by solar energy as a solution.
- The technology enables charging without physical connections, improving safety, accessibility, and convenience. By harnessing solar power, it reduces dependence on traditional power sources, making the system both cost-effective and environmentally friendly. Additionally, the system allows EVs to charge while in motion, further minimizing downtime.
- This innovative solution addresses the shortcomings of traditional EV charging methods and promotes sustainable, eco-friendly transportation, while contributing to a more flexible and interconnected energy ecosystem for the future.

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- Literature Survey
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INTRODUCTION

- The Solar Wireless Electric Vehicle Charging System is an innovative solution that combines solar energy with wireless power transfer (WPT) technology to offer a sustainable, efficient, and convenient method of charging electric vehicles (EVs).

- This system eliminates the need for traditional wired connections by using electromagnetic fields to transfer energy from a stationary transmitter to a receiver in the vehicle. By harnessing solar power, the system provides a renewable, eco-friendly source of energy, reducing dependency on the grid and promoting the use of clean energy.

- This mini-project aims to design and demonstrate the feasibility of a solar-powered wireless charging system for EVs, contributing to the future of sustainable transportation.

LITERATURE SURVEY

Sl no	Paper name	Author name	Insights	Date of publishing
1	Design of an Inductive Power Transfer System for Solar-Powered Electric Vehicle Charging.	Liu Y etal	• The paper presents a design methodology for an inductive power transfer (IPT) system that combines solar energy to wirelessly charge electric vehicles. • It emphasizes system efficiency, power transmission and control stratergies.	[2024]
2	Design and Analysis of a Solar-Powered Wireless Charging System for Electric Vehicles.	Agarwal S	• The paper discusses the design and performance analysis of a solar-powered wireless charging system for EVs. • It provides a thorough examination of power conversion and energy transmission methods.	[2023]
3	Advancements in Wireless Power Transfer Systems for Electric Vehicle Charging.	Bose B	• The paper includes basic knowledge and information regarding uart and focuses on topics like uart parameters, data framing and more. • From the paper we have derived a better understanding on the workings and fundamentals of UART.	[2022]

GAPS IDENTIFIED

- **Limited Power Output:** Wireless power transfer for EV charging typically requires substantial power, which might exceed the capabilities of a mini-project's power supply.
- **Power Storage (Batteries OR Capacitors):** Storing the solar energy for use during non-sunny periods or to stabilize energy flow can be challenging in small-scale systems.
- **Size of the Solar Panel:** The size of the solar panel in our mini-project may not generate enough power to charge an EV, especially if you are targeting even small electric vehicles or models.

OBJECTIVES

- The objective of a solar wireless electric vehicle (EV) charging system is to provide a sustainable, convenient, and efficient charging solution by using solar energy to wirelessly charge EVs.

- This system aims to reduce reliance on the grid, lower carbon emissions, minimize maintenance, and enhance user convenience through wireless charging, all while advancing renewable energy technologies and supporting energy independence.

METHODOLOGY

Phase 1: Literature Review:

1. Research existing studies on solar-powered wireless electric vehicle charging stations.
2. Review the current state of technology in wireless charging and solar power.
3. Identify the key components and design considerations for the charging station.

Phase 2: System Design:

1. Design the solar panel array and calculate the required power output.
2. Select a suitable wireless charging technology and design the charging coil.
3. Design the power electronics and control systems for the charging station.

Phase 3: Hardware Implementation:

1. Procure the necessary components, including solar panels, wireless charging coils.
2. Assemble and test the solar panel array and wireless charging coil.
3. Implement the power electronics and control systems for the charging station.

Phase 4: Testing and Evaluation:

1. Test the charging station's efficiency and effectiveness.
2. Evaluate the environmental impact of the charging station.
3. Identify areas for improvement and optimize the charging station's performance.

METHODOLOGY

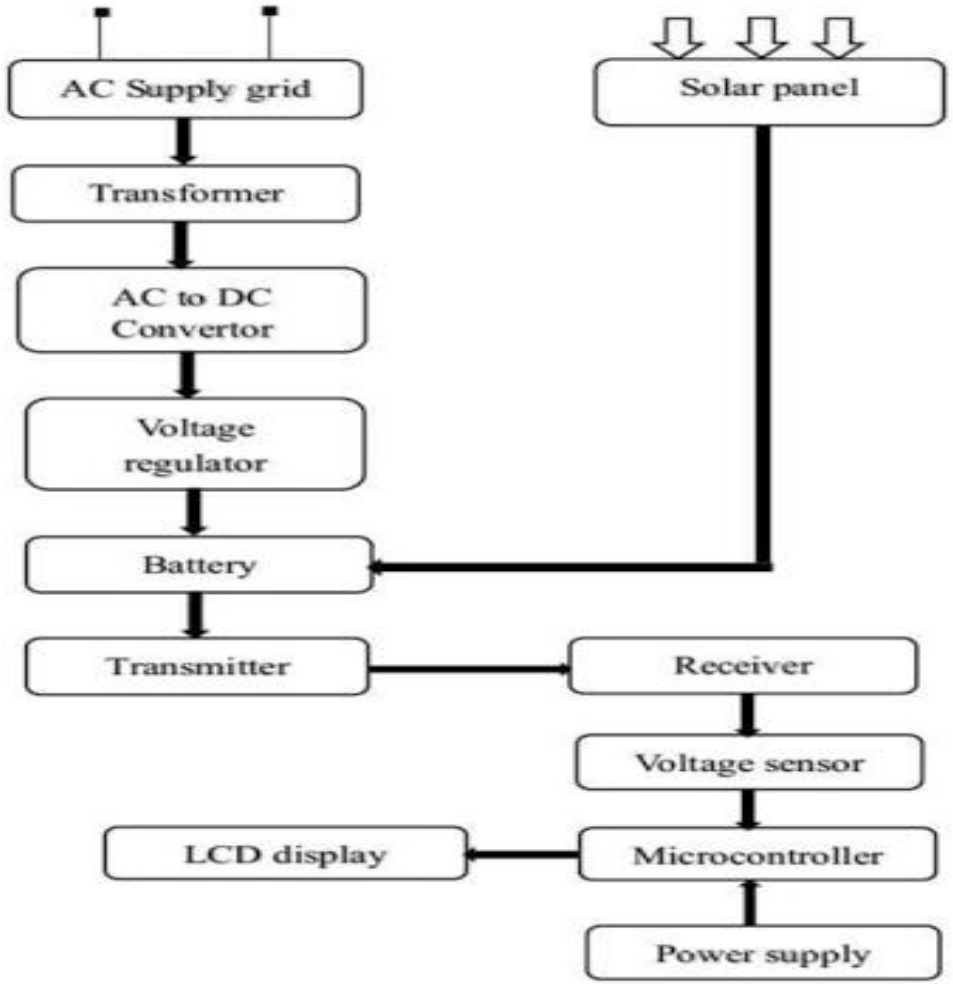


Fig: Block diagram of Solar Wireless Electric Vehicle Charging system

❖ Hardware Implementation:

1. **Solar Panel Array:** Install a solar panel array with a capacity of 1-2 kW.

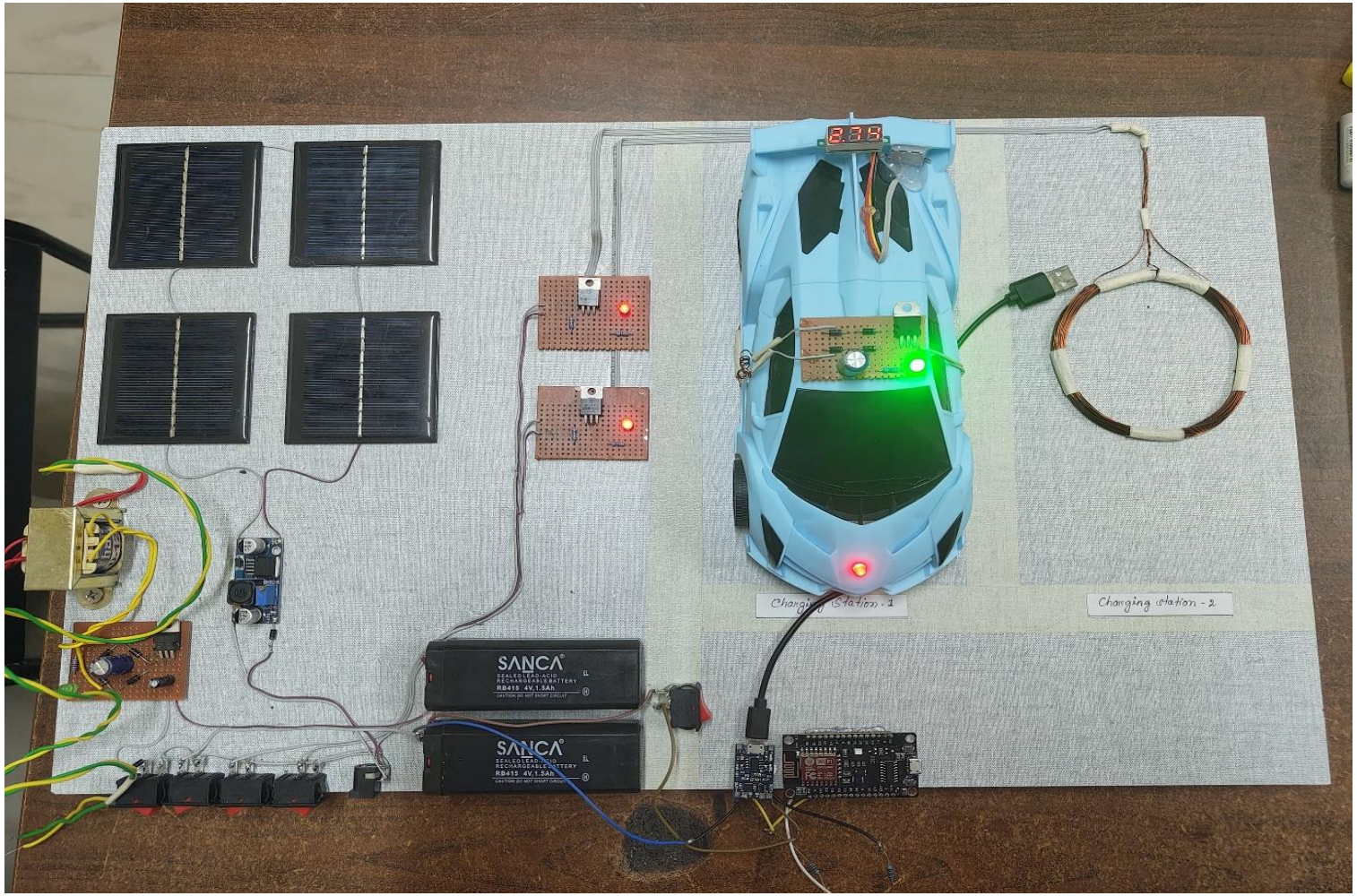
2. **Wireless Charging Pad:** Install a wireless charging pad with a power output of 1-2 kW.

3. **Power Conditioning System (PCS):** Install a PCS to convert DC power from the solar panels to AC power for charging the electric vehicle.

4. **Electric Vehicle (EV) Charging System:** Install an EV charging system to receive and convert the wireless energy into DC power for battery charging.

5. **Monitoring and Control System:** Install a monitoring and control system to track the performance of the solar panel array, wireless charging pad, and PCS.

❖ . DESIGN OF SOLAR WIRELESS ELECTRIC VEHICLE CHARGING SYSTEM:



❖ CALCULATIONS:

Given Data:

- Battery Voltage (V): 5V
- Battery Capacity (mAh): 2000mAh (Example)
- Solar Panel Output (W): 10W, 5V.
- Charging Efficiency: 85% (Wireless Power Transfer).

1. Energy Required to Fully Charge the Battery:

- Energy required by the battery:
- Energy=Voltage×Capacity
- Energy=5V×2000mAh=10,000mWH=10Wh

2. Energy Delivered by the Solar Panel:

- Solar panel provides 10W.
- Assuming 4 hours of optimal sunlight per day:
- Energy Generated=Solar Power×Sunlight Duration=10W×4h=40Wh
- Accounting for 85% efficiency during wireless power transfer:
- Usable Energy=40Wh×0.85=34Wh

RESULTS & OBSERVATIONS

3. Time to Fully Charge the Battery:

The charging time is calculated as:

- Charging Time= Energy Required/Power Delivered
- Charging Time=10Wh/10W×0.85=10/8.5≈1.18hours(1 hour 11 minutes)

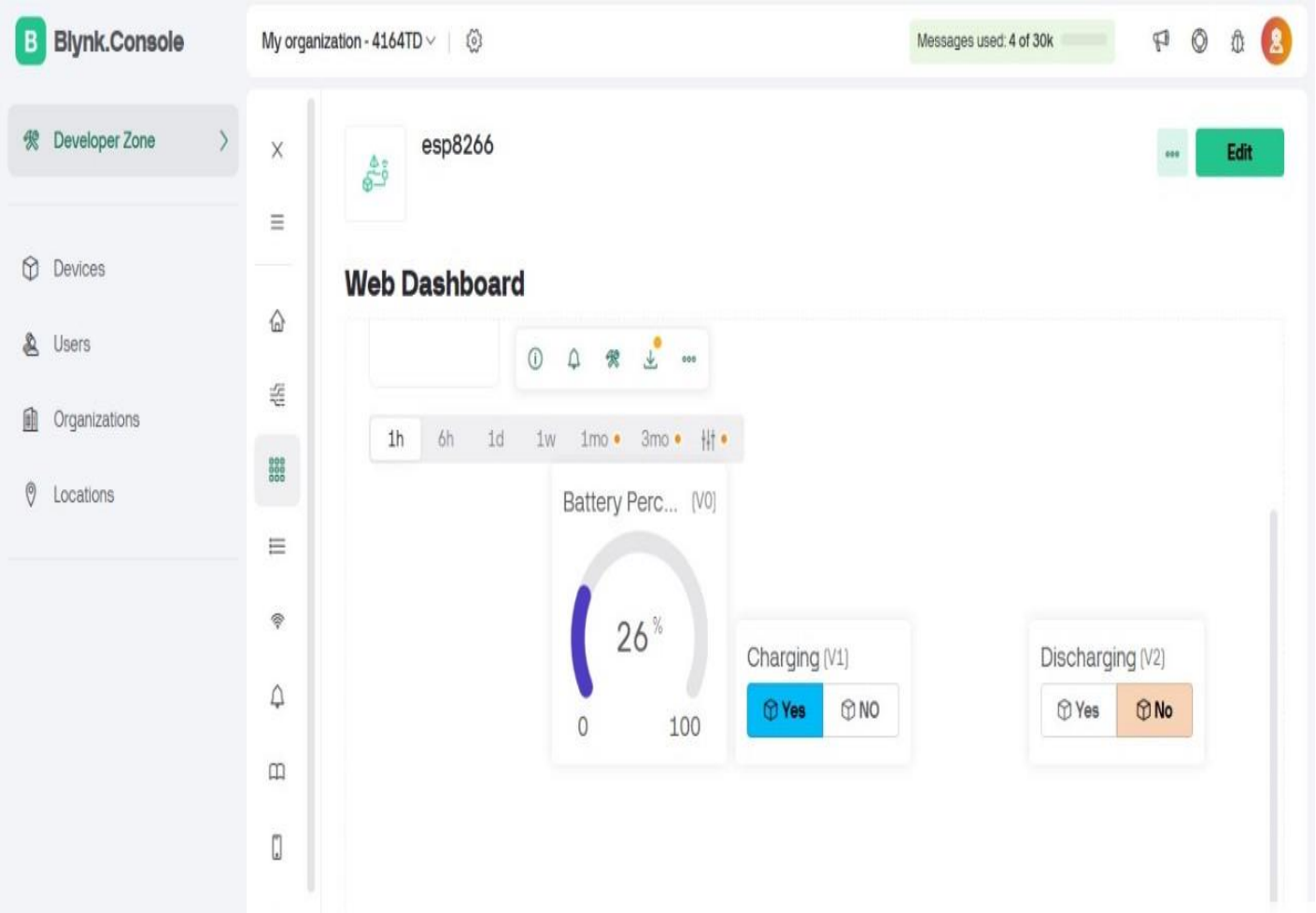
4. Battery Percentage Increment Per Hour:

- Increment Rate=Battery Capacity Charging Power×100
- Increment Rate=10Wh8.5W×100=85%per hour
- Thus, the battery charges at a rate of 85% per hour, reaching 100% in approximately 1.18 hours.

❖ **<u>Key Results:</u>**

Time to Fully Charge: ~1 hour 11 minutes.
Energy Generated by Solar Panel: 40Wh/day.
Usable Energy for Charging: 34Wh/day (after accounting for wireless efficiency).
Real-Time Battery Percentage Increment: ~85% per hour.

❖ RESULTS DISPLAYING ON WEB DASHBOARD IN BLYNK IOT:



❖ RESULTS:

The outcome of this solar wireless EV charging system is that it will provide an efficient, sustainable, and user-friendly solution for EV charging. The system is anticipated to:

- ❖ Optimize energy efficiency using wireless power transfer.
- ❖ Contribute to environmental sustainability by utilizing solar energy.
- ❖ Offer greater convenience with wireless charging and automatic alignment.

CONCLUSION

- The Solar Wireless Electric Vehicle (EV) Charging System represents an innovative and eco-friendly solution to the growing demand for sustainable transportation. By harnessing solar energy and utilizing wireless power transfer (WPT) technology, the system provides a convenient, efficient, and clean method for charging electric vehicles without the need for cables.

- In conclusion, the Solar Wireless EV Charging System offers a future-proof, sustainable solution that can play a vital role in the adoption of electric vehicles, reduce environmental impact, and promote energy efficiency. This miniproject demonstrates the potential of combining renewable energy with modern wireless technologies to meet the growing needs of green transportation.

FUTURE WORK

- **Future Directions Bidirectional Charging:** Enabling the vehicle to return power to the grid or house (Vehicle-to-Grid, V2G) can create a more efficient energy ecosystem.

- **Improved Materials:** Advances in coil materials and design can further improve the efficiency of wireless energy transfer.

- **Smart Charging Networks:** Integrating these systems with smart grids and IoT devices could enable intelligent charging based on grid demands, solar generation, and vehicle energy needs..

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